SAMUEL HINN LIONEL KATAM

DATA SCIENTIST | CLUSTERING & PREDICTIVE MODELING

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PROFILE

Data Science MSc graduate with applied research experience in hierarchical clusterwise regression, unsupervised learning and spatial data analysis. Built R-based workflows covering cluster formation, cluster-specific linear models, validation and prediction evaluation on Italian emissions data. Seeking a junior data science or analytics consulting role.

TECHNICAL SKILLS

Programming & tools R, RStudio, ggplot2, dplyr, sf, caret and FNN.

Machine learning Hierarchical clustering, density- and centroid-based methods, nearest-neighbour assignment and cluster-specific regression.

Model evaluation Train/validation/test design, model selection, RSS, RMSE and residual analysis; spatial data preparation.

SELECTED ACADEMIC PROJECTS

Master's Thesis | Hierarchical Clusterwise Regression

2025–2026

Università degli Studi della Campania "Luigi Vanvitelli" | Caserta, Italy

- Designed a hierarchical clusterwise regression framework, iteratively merging 106 initial province-based groups using a regression residual-sum-of-squares criterion.
- Analysed spatially referenced 2021 Italian municipal CO_2 , PM2.5 and PM10 data; trained cluster-specific OLS models after area normalisation.
- Assigned validation/test observations to training clusters using a nearest-neighbour approach; compared K = 2–25 on validation RSS (selected K = 2).
- Evaluated held-out predictions with RSS/RMSE and created maps, elbow plots and observed-versus-predicted figures in R.

Bachelor's Thesis | Density-Based vs. Centroid-Based Clustering

2023

Università degli Studi della Campania "Luigi Vanvitelli" | Caserta, Italy

- Investigated density-based versus centroid-based clustering approaches and their application to real datasets.

RELEVANT EXPERIENCE

Data Science Workshop Facilitator | Coding Girls

Mar–May 2023

I.I.S. "Francesco Giordani" | Caserta, Italy

- Explained statistics and data visualisation in RStudio to students aged 14–19, guiding practical exercises and small projects.

EDUCATION

MSc in Data Science

Sep 2023 – Mar 2026

Università degli Studi della Campania "Luigi Vanvitelli" | Caserta, Italy

BSc in Data Analytics

Sep 2019 – Jul 2023

Università degli Studi della Campania "Luigi Vanvitelli" | Caserta, Italy

ADDITIONAL TRAINING

Python programming workshop (introductory); additional coursework in C, C++ and Java.

LANGUAGES

English — advanced (self-assessed) | Italian — conversational | Telugu — native